

Mitigating AI Bias Recognition to Enhance Accuracy for South East Asian Features in Immigration Sector

Rachel Marvin Wan

Course: Multimedia Computing
Faculty of Computer Science and Information Technology
Sarawak, Malaysia
81010@siswa.unimas.my;
rachelmarvin2002@gmail.com

AP Dr. Jacey Lynn- Minoi

Faculty of Computer Science and Information Technology
Universiti Malaysia Sarawak (UNIMAS)
Sarawak, Malaysia
jacey@unimas.my

Abstract—Facial recognition systems used in immigration sectors often exhibit lower performance on under represented populations, particularly individuals from Southeast Asia. This project presents a prototype facial recognition pipeline tailored to Southeast Asian faces with the aim of mitigating demographic bias and improving identification reliability for this region. The system integrates Multi Task Cascaded Convolutional Neural Network (MTCNN) for face detection and alignment, an InceptionResnetV1 (FaceNet) model for deep embedding extraction, and a supervised classifier for four nationality classes: Bruneian, Indonesian, Malaysian and Thai. A curated dataset of Southeast Asian facial images was organised using a structured directory layout and augmented through controlled transformations such as rotation, brightness and saturation adjustment to increase variability while preserving key facial characteristics. A standardised preprocessing pipeline ensured consistent cropping, resizing and normalisation across training and validation stages. The prototype was evaluated using conventional classification metrics and confusion matrices, with additional per class analysis to examine disparities between nationalities. The results demonstrate the feasibility of building an end to end facial recognition system specifically focused on Southeast Asian populations and highlight persistent challenges related to data limitations and unequal performance across demographic groups. This study provides a practical foundation and methodological baseline for future work on fairness aware facial recognition in regional immigration applications.

Keywords—artificial intelligence bias, facial recognition, immigration technology, machine learning fairness, southeast asian

I. INTRODUCTION

Artificial Intelligence (AI) has been evolving over the years in various sectors, especially in immigration. Under AI itself, several technologies have been developed, and to date, they continue to evolve to ease the process in our daily life activities and continuously improve the lifestyle of our current society. Facial Recognition or Facial Recognition Technology (FRT) are among the widely used AI- based technology in today's life. Implemented in verification systems, surveillance cameras and health-monitoring, it is an AI-based biometric technology that utilizes computer vision to analyze facial images and identify individuals based on their unique facial features (Christian,

2024). Despite it's various use, however the technology itself is found to be bias. Bias could be in terms of gender, age, ethnics, skin colors. In this context, this project focuses on ethnic groups, focusing on South East Asians and other underrepresented group. Studies has found that most facial recognition system are biased against certain ethnic groups (Khalil et al.,2020). This resulted with complications such as high rates of misidentification, inefficient delays and wrongful denial of entrance during the immigration process. Thus, developing bias aware facial recognition system is designed and developed to address these issues.

A. Problem Statement

Facial recognition systems in immigration systems often exhibit bias against certain ethnic groups, particularly those from South East Asia. This bias leads to unfair treatment, longer processing times or wrongful denials of entry. The root causes of this bias include underrepresentation in training datasets, algorithmic limitations in recognizing diverse facial features, and a lack of consideration for ethnic diversity in system design and testing. Addressing this bias is crucial in ensuring equitable and efficient immigration processes. Furthering this issue would not only increase the current problems addressed but could also lead to unjust deportation across borders.

B. Objectives

The main objective of this project is to develop a facial recognition prototype system that could help enhance accuracy for South East Asian and other under-represented ethnic groups using deep learning technique. Other objectives includes:-

- 1.To develop datasets consist of South East Asian and other under-represented ethnic groups features for training and testing purposes.
- 2.To determine the most suitable deep learning techniques and implement them to increase the accuracy of the identification of the South East Asian individuals.
- 3.Evaluate system's performance using fairness and accuracy metrics.

C. Methodology

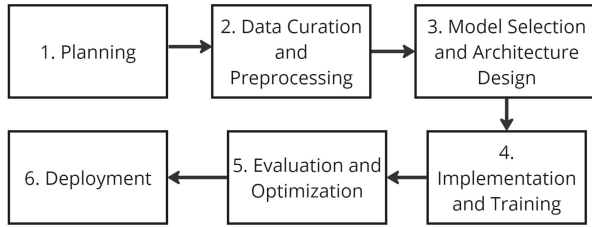


Figure 1.1: Research Methodology

As seen in figure 1.1, the waterfall approach methodology will be used for this project. This approach is used to outline the steps necessary to guarantee that each process is carried out correctly. The primary goal of the planning stage will be to determine the resources required for this undertaking. This entails planning the entire system, defining the goals precisely, and locating the datasets from reliable open-source platforms like Kaggle. Prior to being trained on a dataset, datasets will be gathered from the designated sources and categorized by gender, age ranges, and ethnic groups utilizing preprocessing techniques, such as scaling and normalization as needed using Roboflow. The planning phase's prototype system will then be developed and detailed as necessary. Following this phase, the prototype will be put into use and tested to replicate various immigration circumstances. Fairness and correctness will then be assessed. In the implementation phase, the proposed system is realised as a modular facial recognition pipeline combining dataset preparation, face detection, embedding extraction and classification. Raw facial images are organised into training and validation splits using a YOLO-style directory structure, and data augmentation such as rotation, brightness and saturation adjustments is applied in Roboflow to increase diversity while preserving key facial characteristics. During preprocessing, the Multi-Task Cascaded Convolutional Neural Network (MTCNN) detects and aligns faces, and the cropped regions are saved into dedicated folders for reuse. A pretrained InceptionResnetV1 (FaceNet) model is then used to generate 512-dimensional embeddings for each cropped face, and a supervised classification layer is trained on these embeddings to predict the target nationality class. The same pipeline is applied during inference, and model performance is assessed using accuracy, confusion matrices and per-class precision, recall and F1-scores.

D. Scope

The significance of this project lies in its potential to reduce racial bias in AI applications, particularly in face recognition systems used in the immigration industry. Currently, the accuracy of AI systems for members of underrepresented groups, such as Southeast Asians, accurately predicts their traits. This project aims to bridge that gap by creating a prototype specifically designed to accurately recognize Southeast Asian features.

The suggested model helps to develop fair and ethical AI by fostering inclusivity in an area where bias can have serious

effects, such as delays, increased scrutiny, or wrong denials in immigration processes. This effort may have an impact on the creation of future systems that cater to multicultural populations by highlighting the significance of data variety and algorithmic adaptation. Furthermore, this prototype could serve as a foundational example for other sectors using facial recognition technology, such as security, healthcare, and employment, offering a pathway to develop fairer AI-driven systems.

E. Significance of the Project

The significance of this project lies in its potential to lessen racial bias in AI applications, especially in the face recognition systems used within the immigration industry. Currently, the accuracy of AI systems for members of under-represented groups, such as Southeast Asians, traits accurately. This project aims to bridge that gap by creating a prototype specifically designed to recognize accurately recognized Southeast Asian features.

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F. Expected Outcome

The expected outcome of this project include:-

1. Increase of accuracy and fairness - Significant improvement of a facial recognition system designed to recognize Southeast Asian traits with better accuracy and fairness compared to traditional facial recognition models.
2. Data diversity insights - A curated and representative datasets highlighting the importance of diversity in training data for AI system.
3. Development of prototype system - A working prototype of a facial recognition system that would recognize Southeast Asian facial features with reduced bias.

II. LITERATURE REVIEW

A. Type of Bias

Ethnic bias

Facial recognition systems perform worst on Southeast Asian faces compared to Western. A study found by Buntoun and Kuok(2023) found that systems trained on Western datasets showed significant performance drops when tested on Southeast Asian faces. The systems frequently confused

different Southeast Asian ethnicities, misclassifying Cambodian individuals as Vietnamese and vice versa. This ethnic bias creates real problems in immigration. Southeast Asian travelers experience higher rates of false rejections, requiring manual verification that causes delays and increased scrutiny. In worst cases, this leads to wrongful denial of entry.

Data Training Bias

Most facial recognition datasets severely underrepresent Southeast Asian faces. In 13 major face databases are reviewed and found they primarily contain Western and East Asian faces, with Southeast Asian faces either missing or minimally represented (Khalil et al., 2020)

The largest training datasets like MS-Celeb-1M (10 million images) and VGGFace2 (3.3 million images) come from celebrity databases that reflect Western entertainment demographics. When these datasets include "Asian" categories, they typically mean East Asian, not Southeast Asian faces. This data gap means algorithms never properly learn Southeast Asian facial features, creating systems that work well for some populations but fail for Southeast Asians.

Cross-race Bias

Facial recognition algorithms work better on faces similar to their training data. Phillips et al. (2011) showed that Western-developed algorithms perform better on Western faces, while East Asian-developed algorithms perform better on East Asian faces. Southeast Asians get caught in the middle - they're not well-represented in either Western or East Asian training datasets. This creates a "double disadvantage" where no existing systems are optimized for Southeast Asian facial characteristics.

System Bias

Recognition systems employ universal decision thresholds to ascertain matches; nevertheless, these thresholds function variably among different ethnic groups. Using the same threshold for everyone creates unequal error rates across different populations (Robinson et al., 2020). For Southeast Asians, this implies that the system may be excessively stringent (resulting in erroneous rejections) or overly permissive (leading to security vulnerabilities). In immigration circumstances where precision and security are paramount, these threshold issues pose significant operational challenges.

B. Related Works

Several research work that developed this solution are increasing drastically and different methods are implemented in mitigating this issue. One of the fundamental techniques in facial recognition is Principal Component Analysis, also referred to as the "Eigenfaces" method (Hu et al., 2017). PCA preserves the most significant variance in the data while reducing high-dimensional face image data to a lower-dimensional space. The technique converts photos of faces into eigenfaces, which are basic elements that represent the main differences between various faces. Studies that exhibit significant feature dimension reductions while preserving discriminative power attest to PCA's efficacy in dimension

reduction (Islam et al., 2018). One approach maintained a 99% variance ratio while achieving a 99.72% dimension reduction, lowering features from 69,240 to just 191. PCA's inability to handle changes in illumination and its emphasis on global rather than local facial traits, however, can affect recognition accuracy in difficult situations (Majumder & Bhowmik, 2015). PCA's key flaw is that it optimizes for variance rather than class separability, despite the fact that it is still theoretically beautiful and computationally efficient. This implies that when within-class differences outweigh between-class differences, it might not be able to differentiate between people efficiently. Using Linear Discriminant Analysis or LDA is another solution that is developed in mitigating this issue. Sometimes being referred to as "Fisherface", it is very useful for recognition tasks when differentiating between identities is essential since it looks for projection axes that maximize between-class variance while minimizing within-class variance. This is specially good especially it optimized class separability rather than just just variance, which overcomes some of PCA's drawbacks (Hu et al., 2017). By establishing feature spaces where photos of the same person cluster closely together but images of different people stay well apart, the study demonstrates the utility of LDA in improving feature discrimination (Santhosh & Rajashekararadhya, 2023). Because of this characteristic, LDA is particularly helpful for verification tasks in immigration contexts where the algorithm has to decide if two photos depict the same individual. When feature dimensions are greater than training samples, LDA's primary drawback is revealed by limited sample sizes, necessitating regularization techniques for stability. Notwithstanding these limitations, LDA is still useful when paired with other techniques to improve system performance. Encouragement For facial categorization tasks, vector machines have proven to be incredibly effective; several research have shown that they perform better than conventional techniques (Islam et al., 2018; Majumder & Bhowmik, 2015). SVM is especially well-suited for high-dimensional facial recognition problems because it finds the best hyperplane in feature space that optimizes the margin between classes. When paired with suitable feature extraction techniques, research by Islam et al. (2018) showed that SVM is successful in facial expression recognition, attaining accuracies of above 95%. According to the study, SVM performs well in high-dimensional spaces and manages scenarios when feature counts are more than sample counts, which is a typical occurrence in facial recognition. In comparison to more sophisticated kernels, Majumder & Bhowmik (2015) demonstrated the computational efficiency of linear kernel SVM with thermal facial recognition by achieving recognition rates above 99%. SVM's mathematical underpinnings and capacity to manage intricate decision boundaries are its main advantages. However, because the method is binary, it must be modified for multi-class situations using strategies like One-Versus-One or One-Versus-Rest, which might make computation more difficult. Additionally, parameter tweaking and kernel selection have a significant impact on SVM performance. A conceptually straightforward method for classifying faces is the k-Nearest Neighbors algorithm, which uses the majority class of the k nearest training samples in feature space (Islam et al., 2018). Research shows that k-NN has practical limits when compared to other classifiers, especially with regard to memory needs and

computational efficiency, even though it can reach good recognition results. Since the method needs to keep all training data for classification, k-NN's memory requirements are its primary drawback. This storage need becomes unaffordable for large-scale immigration systems with thousands of enrolled persons. Furthermore, the choice of k and distance measure has a significant impact on performance, therefore choosing the right parameters is essential for best results. Majumder & Bhowmik (2015) introduced a novel hybrid method that combines Fast Independent Component Analysis (FastICA) for dimensionality reduction with Gabor filters for feature extraction. By utilizing the complementing advantages of both approaches, this strategy was able to attain recognition rates on thermal face photos that were higher than 96%. Gabor filters produce 40 filtered versions of each input image in five scales and eight orientations, demonstrating their exceptional ability to capture local texture information at various scales and orientations. But doing so produces high-dimensional data (276,960 features per image), which would be too expensive to process for straightforward classification. In contrast to PCA's variance-based method, FastICA provides more discriminative representations by removing statistically independent components. The Gabor-FastICA method shows how strong recognition systems may be produced by fusing cutting-edge statistical methods with conventional feature extraction. The method's resilience is demonstrated by its performance with difficult thermal images, but real-time applications may be limited by the computing burden of creating and processing many Gabor filters. According to several research, a particularly successful hybrid strategy is the combination of SVM for classification and PCA for dimensionality reduction (Islam et al., 2018). This technique takes advantage of SVM's strong classification skills while utilizing PCA's effectiveness in compressing facial data. According to research, this hybrid approach achieved a 95.7% accuracy rate in facial expression recognition, indicating its efficacy for real-time applications where dimensionality reduction increases computational efficiency and allows for faster processing without sacrificing recognition accuracy. This hybrid approach demonstrates the benefits of using complementary techniques to tackle the intricate problems of facial recognition. These techniques frequently outperform single-method systems by combining the advantages of conventional feature extraction with those of contemporary classifiers, especially when handling varied populations or difficult imaging scenarios. According to the study, facial recognition systems can benefit greatly from both hybrid and classic machine learning techniques. While hybrid approaches frequently offer the optimum mix between performance and processing requirements by combining different techniques, traditional methods like as PCA and LDA continue to be relevant because to their computing efficiency and interpretability. The best ways to mitigate demographic bias in Southeast Asia may be hybrid strategies that combine many feature extraction techniques with adaptive classification algorithms. Hybrid approaches are especially well-suited for enhancing recognition accuracy across underrepresented populations, such as Southeast Asians, because they may implement bias-aware classification algorithms and capture a variety of facial traits using several feature extraction methods. Evaluation frameworks that specifically measure performance

across various ethnic groups are necessary, especially for disadvantaged populations in immigration scenarios, as the majority of research concentrates on overall accuracy rather than demographic fairness.

C. Bias Mitigation Techniques

The methodology incorporates multiple complementary bias mitigation techniques including dynamic loss weighting that automatically adjusts the importance of different training objectives based on current performance across demographic groups. This adaptive approach ensures marginalized populations receive adequate attention during training without compromising overall system effectiveness.

Attention mechanisms specifically designed for Southeast Asian facial features provide an additional dimension of bias mitigation by enabling the model to focus on discriminative characteristics relevant to all target populations. These mechanisms are designed to identify and emphasize features that facilitate accurate recognition while avoiding reliance on demographic shortcuts that could lead to biased outcomes.

D. Evaluation Framework

The evaluation methodology functions as an integrated assessment framework rather than a separate research objective. Bias measurements inform dataset aggregation and balancing procedures during dataset development. Fairness metrics guide training methodologies and architectural choices during algorithm implementation. Bias monitoring capabilities are incorporated into system design during prototype development.

Bootstrap confidence intervals assess uncertainty in bias metrics and provide robust estimates of system performance across different demographic groups.

III. RESULTS

A. Dataset Development

A curated dataset of Southeast Asian faces was constructed by combining images from open-source repositories and event photographs, with a focus on Bruneian, Indonesian, Malaysian and Thai subjects. Images were organised into a YOLO-style directory structure with separate folders per nationality and split into 500 training and 100 validation images. Roboflow was used to apply controlled augmentations, including small rotations and moderate brightness and saturation changes, to increase variability without distorting facial geometry.

B. Model Implementation

The implemented model uses MTCNN for face detection and alignment followed by an InceptionResnetV1 (FaceNet) backbone pretrained on VGGFace2. Detected face crops are resized to 224×224 and standardised before being passed through FaceNet to produce 512-dimensional embeddings. A new fully connected layer is trained on these embeddings to classify each face into one of the four target nationalities. Training is carried out in Google Colab using the Adam optimizer with a learning rate of 0.001, a MultiStepLR scheduler and mini-batches of cropped faces.

C. Experimental Evaluation

Evaluation on the 100-image validation set uses overall accuracy, confusion matrices and per-class precision, recall and F1-scores to analyse performance across nationalities. While overall accuracy remains modest due to the small dataset and high similarity between neighbouring Southeast Asian nationalities, the confusion matrix shows higher recall for Thai faces and lower recall for Malaysian and Bruneian faces, indicating residual demographic bias in the prototype. These results highlight both the feasibility of the MTCNN–FaceNet pipeline for Southeast Asian faces and the limitations imposed by data scarcity and class

Confusion Matrix (Validation Set)

True label \ Predicted label	bruneian_face	indonesian_face	malaysian_face	thai_face
bruneian_face	7	6	0	13
indonesian_face	2	6	1	14
malaysian_face	1	2	6	19
thai_face	1	4	1	17

imbalance.

Figure 1.2: Confusion Matrix for four-class nationality classification.

IV. DISCUSSION

The research successfully demonstrates the feasibility of developing facial recognition prototype that focuses specifically on Southeast Asian populations by combining targeted dataset development with a FaceNet-based pipeline. The implemented system uses MTCNN for face detection and alignment, followed by InceptionResnetV1 (FaceNet) embeddings and a supervised four-class classifier, which together form a practical baseline for analysing nationality-related performance and bias. The curated dataset and Roboflow preprocessing help address the underrepresentation of Southeast Asian faces in existing large-scale benchmarks by providing more realistic samples from Bruneian, Indonesian, Malaysian and Thai individuals.

Despite these contributions, the experimental evaluation highlights several persistent challenges. The relatively small dataset and high visual similarity between neighbouring Southeast Asian nationalities limit the overall recognition performance and lead to noticeable differences in recall across classes, with Thai faces generally recognised better than Malaysian and Bruneian faces. This uneven per-class behaviour indicates residual demographic bias and suggests that the model can still overfit to patterns that are more common in some nationalities than others. The findings therefore reinforce that improving fairness in facial recognition cannot rely solely on model choice; it also requires larger, better-balanced datasets and evaluation protocols that explicitly report group-wise performance rather than only overall accuracy.

V. CONCLUSION

This work presents a FaceNet-based facial recognition prototype aimed at mitigating bias for Southeast Asian populations in immigration-style scenarios. By integrating MTCNN face detection, InceptionResnetV1 embeddings and a supervised four-class nationality classifier trained on a curated Southeast Asian dataset, the study establishes a complete end-to-end pipeline that can be used to investigate both performance and fairness for Bruneian, Indonesian, Malaysian and Thai faces. The results show that while the prototype can correctly classify a portion of validation samples, substantial confusion between nationalities and uneven per-class metrics remain, underscoring the difficulty of nationality prediction from facial appearance alone under limited data conditions.

The prototype therefore serves as a foundational step rather than a deployable system, providing empirical evidence of where current methods fall short for Southeast Asian populations and offering a concrete baseline for future improvements. Future work should focus on expanding and rebalancing the dataset, exploring alternative training strategies such as metric-learning losses or different classifier heads, and incorporating more comprehensive fairness metrics and cross-dataset evaluations. In this way, the approach developed here can be extended towards more reliable and equitable facial recognition systems for immigration and other high-impact applications involving underrepresented demographic groups.

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